Effective Design of Tangible Physical User Interfaces for Entertainment Devices

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***Abstract*—This project explores the design of intuitive physical user interfaces through the development of a tangible bomb-defusal game. Creating intuitive physical user interfaces for operating electronic devices is a challenge that all devices must overcome. Poorly designed interfaces can leave users frustrated and confused, often leading to increased user error and frustration. The goal of this project is to create a device that is easy to understand for the user without the need for instructions on how to operate it. This will be achieved by researching, adopting, and iterating using design techniques and methods commonly used to create intuitive Tangible User Interfaces (TUIs) in devices such as toys and appliances. During development, the importance of leveraging physical affordances and clear tactile feedback became apparent. A digital prototype was developed to test the intuitiveness of the device, allowing rapid prototyping before physical components were assembled. To conclude, iteration and user testing are required to ensure that this device is understandable and accessible.**

Keywords— Physical User Interface, Tactile, Tangible User Interface, Intuitive, Affordances

I. Introduction

In today's gaming landscape, user interaction is increasingly standardized by major platform holders, specifically *PlayStation*, *Xbox*, and traditional *PC* peripherals [1]. While standardized controls ensure accessibility, it often forces game mechanics to conform to established design conventions, limiting a game's potential for unconventional, tactile gameplay. This project aims to address the lack of physical immersion by developing a standalone game housed within a controller. Inspired by the unconventional gameplay of *Keep Talking and Nobody Explode*s [2] and *Bop It* [3], this controller will adapt these gameplay concepts to a unique physical controller emphasizing tactile manipulation, in the form of a bomb-defusal game featuring multiple mini-games.

The challenge this design faces is maintaining player engagement and consistently invoking a sense of “thrill” using a physical experience. However, this problem is not unique to our design. Arcade manufacturers, casino game designers, and toy manufacturers must capture the same sense of thrill, and also must do so by bridging the gap between a user and a virtual world using a mechanical system. Current industry solutions often focus on haptic feedback and multi-sensory visual or auditory cues, such as flashing lights, vibration, or sound, to maintain "flow state" [4].

Failing to provide meaningful tactile feedback will lead to a lack of interest in the user and may also create confusion. In an increasingly competitive entertainment market, products must compete for the user's attention span. Failing to communicate a product's state correctly may lead to diminished user retention, pushing them to use other products instead. By integrating game logic directly into the controller, this project aims to demonstrate that tactile-centric design can foster a deeper connection between the user and the interactive experience, ensuring this device remains intuitive [5].

II. Background Review

To identify literature relevant to this project, the following search terms were used: Physical User Interface, Intuitive Physical User Interface, Tangible User Interface (TUI), and Tangible User Interface Toys. These terms were chosen as they highlight user interfaces designed with intuitiveness, physicality, tangibility, and play in mind.

Based on initial results, six sources were discarded. Three sources were general purpose books, meaning they were less concise than academic research papers. The other three discarded sources lacked direct relevance to this project's research goals.

Research on TUIs is a relatively small field of study. Citations [7], [8], and [11] expand on the last in chronological order. [7] and [8] identify a lack of formalization for the development of TUIs, both proposing frameworks to design and iterate on TUIs. [11] makes reference to the other two papers and aims to create a roadmap to help developers by compiling and reviewing 64 studies. The authors categorize TUIs into Sensory (internal sensors) and Computer Vision (cameras/markers) approaches. The takeaway is that object recognition remains the main bottleneck for widespread TUI adoption. For this project, sensors must be accurate to provide responsive feedback.

Next, citations [6], [9], and [10] identify that while traditional Graphics User Interfaces (GUIs) hinder dexterity and require "encoding" of new knowledge, TUIs leverage

physical “affordances” allowing prior knowledge to factor into intuitability. Specifically, [6] highlights how grip and thumb position dictate usability, while [9] and [10] demonstrate that tactile interfaces significantly increase engagement and intuitiveness, especially for children. Together, these sources can be used to infer that for this project to succeed, ergonomics and immediate tactile feedback should be prioritized over digital menus. By focusing on TUIs over GUIs, a seamless and low-latency user experience can be achieved.

Lastly, Citations [12] and [13] highlight usability in Simon and Bop-it. As TUIs, both bridge physical manipulation and digital feedback, leveraging tactile interaction over screens. [12] notes Simon's engagement stems from tactile buttons and novelty, while [13] emphasizes how Bop-it trains working memory and motor skills through challenging physical actions. Ultimately, research suggests intuitive TUIs must react via lights, sounds, or feel, keeping the user focused on the physical interaction.

III. Results

The initial project concept emerged from a design thinking session. Preliminary designs included an “interactive house” device, PC-compatible motion controllers, and a haptic "bomb-defusal" game. Ultimately, the bomb-defusal controller was selected for its mechanical novelty. To maximize user engagement, the controller was designed to "thrill" the user. A comprehensive heavyweight use-case analysis was performed to validate the selection of integrated mini-games, as detailed in Appendix B.

Additional design thinking sessions were conducted with an external peer group to evaluate controller ergonomics and mini-game designs. Furthermore, independent conversations with peers led to the ideation of additional elements to raise the thrill factor of this device. The feedback from these sessions was adopted into a Quality Function Deployment (QFD) matrix. The QFD analysis shown in Appendix C prioritized "intuitiveness" and "uniqueness" as the primary engineering characteristics. This analysis was considered during the transition from a cardboard prototype to a functional controller.

Based on priorities outlined in the QFD matrix, system architecture was developed to handle user input and output. As illustrated in Appendix D, the digital diagram of the controller utilizes an Arduino microcontroller to interpret user input and trigger corresponding actuators. To satisfy the requirements for a group of two, the digital diagram incorporates two sensors and two actuators. For the sensors, a tilt sensor and a matrix keypad were used. The two actuators are red and green LED lights.

The physical hardware, demonstrated in Appendix E, replaces the tilt sensor with an accelerometer for precision motion sensing and utilizes a Liquid Crystal Display (LCD) to replace one of the LEDs to provide dynamic status feedback. The physical controller retains the matrix keypad one primary LED. The selection of components are detailed in the Bill of Materials (BOM) provided in Appendix F.

IV. Conclusion

The initial research and development of this bomb-defusal controller highlighted the importance of novelty and responsive feedback while developing effective TUIs. By leveraging physical affordances such as motion sensing via an accelerometer and direct feedback via an LCD and LED, this controller bridges the gap between physical manipulation and digital interaction. Going forward, future iterations will adopt a more constrained approach to the design, with considerations for the technical limitations identified during this phase of development. As documented in the timeline (Appendix G), these finalized steps will be executed to complete the physical prototype.

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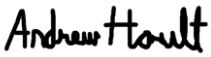
APPENDIX A

Name: Andrew Hoult
Role: Hardware, Design

Contributions:

- I assisted with documentation
- I handled the physical hardware
- I helped design and ideate the controller



Name: Ethan Schledewitz-Edwards
Role: Documentation, Programming, Design
Contributions:

- I handled the majority of documentation
- I developed the digital Unity prototype
- I modeled and rendered the 3D blender prototype
- I helped design and ideate the controller

APPENDIX B

Use Case 1	Beat the wire game before the time runs out
Actor	The Player
Basic Flow	1. The player observes that the LED lights around the wires game are illuminated. 2. The user recognizes that each of the 3 LEDs is next to a wire socket 3. The user inserts all three wires into sockets with lit LEDs 4. The user completes the minigame 5. The system gives the user a random digit (0-9) via speaker

Use Case 2	Shake the device before the time runs out
Actor	The Player
Basic Flow	1. The user observes that a panel on the side of the bomb is illuminated 2. The user pulls the panel off 3. The user observes that a light strip fills like a progress bar as the device moves 4. The user shakes the device rapidly 5. The user completes the minigame 6. The system gives the user a random digit (0-9) via speaker

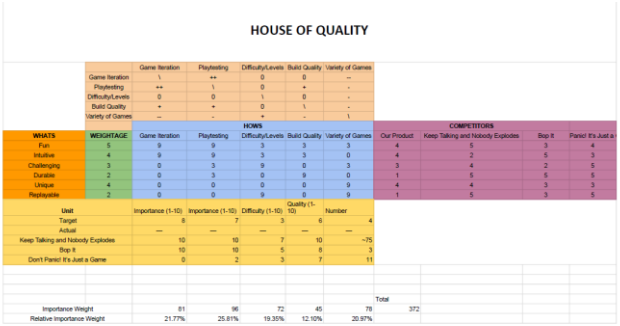
Use Case 3	Play a rhythm game before the time runs out
Actor	The Player
Basic Flow	1. The user observes that the main LCD is lit 2. The user sees a single digit appear on the screen to a specific tone 3. The digit disappears from the screen, forcing the user to remember the digit 4. The user inputs the number to confirm their memory 5. The system prints the original digit followed by a new original digit before they both disappear 6. The user inputs the sequence in the exact order it was presented to continue the cycle 7. The system stops printing numbers after six cycles 8. The user completes the minigame 9. The system gives the user a random digit (0-9) via speaker

Use Case 4	Spin the joystick
Actor	The Player

Basic Flow	1. The user observes LED lights near the battery panel 2. The user observes a vibration near the battery panel 3. The user pulls the battery panel off of the back of the device 4. The player sees a lit LCD screen and a joystick underneath 5. The player flicks the joystick and sees the screen change state 6. The player spins the joystick to fill the LCD panel like a progress bar 7. The user completes the minigame 8. The system gives the user a random digit (0-9) via speaker

Use Case 5	Insert the code
Actor	The Player
Basic Flow	1. The user observes that the main LCD is lit and looks like numbers can be input 2. The user inputs the 4 digits they were previously given after winning each game in sequence 3. The user hits the diffuse button 4. The user wins

APPENDIX C



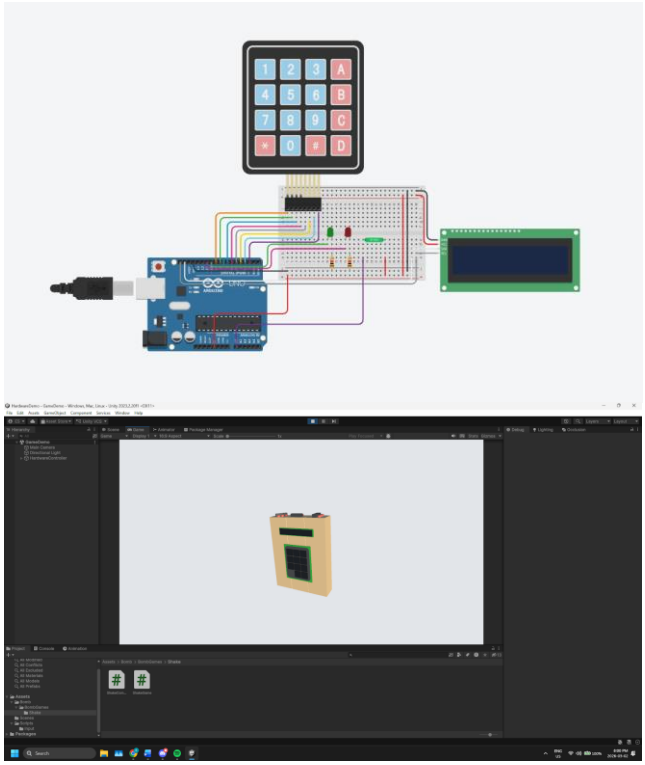
HOUSE OF QUALITY

		Game Iteration	Playtesting	Difficulty/Levels	Build Quality	Variety of Games
	Game Iteration	\	++	0	0	--
	Playtesting	++	\	0	+	-
	Difficulty/Levels	0	0	\	0	-
	Build Quality	+	+	0	\	-
	Variety of Games	--	-	+	-	\

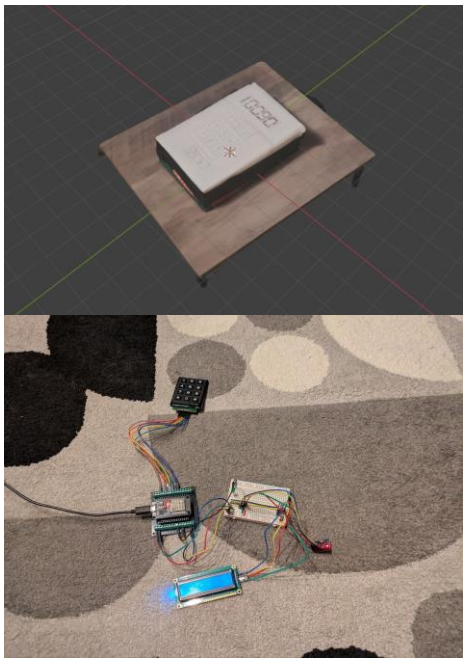
		HOWS					COMPETITORS			
WHATS	WEIGHTAGE	Game Iteration	Playtesting	Difficulty/Levels	Build Quality	Variety of Games	Our Product	Keep Talking and Nobody Explodes	Bop It	Panic! It's Just a
Fun	5	9	9	3	3	3	4	5	3	4
Intuitive	4	9	9	3	3	0	4	2	5	3
Challenging	3	0	3	9	0	3	3	4	2	5
Durable	2	0	3	0	9	0	1	5	5	5
Unique	4	0	0	0	0	9	4	4	3	3
Replayable	2	0	0	9	0	9	1	5	3	5

Unit	Importance (1-10)	Importance (1-10)	Difficulty (1-10)	Quality (1-10)	Number	
Target	8	7	3	6	4	
Actual	---	---	---	---	---	
Keep Talking and Nobody Explodes	10	10	7	10	~75	
Bop It	10	10	5	8	3	
Don't Panic! It's Just a Game	0	2	3	7	11	
					Total	
Importance Weight	81	96	72	45	78	372
Relative Importance Weight	21.77%	25.81%	19.35%	12.10%	20.97%	

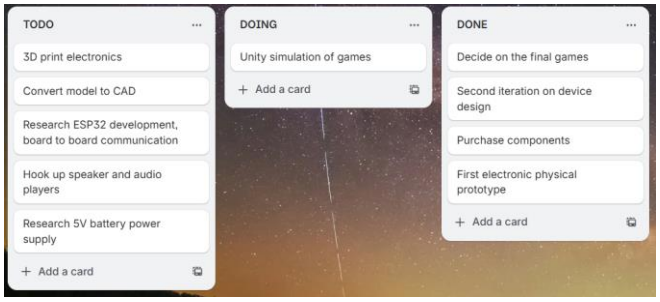
APPENDIX D



APPENDIX E



APPENDIX F

Title	Comments	Price	Quantity	Has
DOITOOL Mini Flat Vibration Motors 10pcs Pack Micro Dc Motor Accessories for Electronics Repair and DIY by DOITOOL (Misc.) |		$8.79	1	0
TUOFENG 22 awg Wire Solid Core Hookup Wires-6 Different Colored Jumper Wire 30ft or 9m Each, 22 Gauge Tinned Copper Wire PVC (OD: 1.60mm) Hook up Wire Kit by TUOFENG |		$23.49	1	0
Beikell SD Card Reader,High-Speed USB C/USB 3.0 Memory Card Reader, 2 in 1 OTG Adapter for SD/Micro SD/MMC/TF/SDXC/SDHC/Micro SDHC Compatible with MacBook Pro Air,iPad Pro Air,Galaxy S21 by Beikell (Personal Computers) |		$10.66	1	0
Patriot LX Series 64GB Micro SDXC Class 10 Memory Card - UHS-I U1 - PSF64GMDC10 by Patriot Memory (Personal Computers) |		$15.99	1	0
3PCS Mini MP3 Player Audio Module with 5PCS Metal Shell Round Internal Magnet Speaker 2W 8Ohm MP3 Voice Decode Board TF Card U Disk IO/Serial Port/AD Board DFPlayer Audio Music Module for Arduino by UMLIFE (Personal Computers) |		$17.53	1	0
Freenove I2C IIC LCD 1602 Module (2 Pack), New Type TWI Serial 16x2 Display, Compatible with Arduino Raspberry Pi Pico ESP32 ESP8266 by FREENOVE (Personal Computers) |		$17.95	1	0
SazkJere 2PCS 3x4 Matrix Keypad 12 Keyboard Module External Keypad Board for Microcontroller (12key) by SazkJere |		$15.99	1	0
CANADUINO® 5 x RGB LED Strip with 8 x WS2812B RGB - Fully addressable by CANADUINO |		$11.99	1	0
Wishiot 5pcs MPU-6050 Acceleration Sensor 6dof Three-Axis Accelerometer Gyro Sensor IMU with 1pc 40Pin Male Header and 1pc 40Pin Male to Female Dupont Cable for Arduino RC Quadcopter Drone by Wishiot (Electronics) |		$19.59	1	0

APPENDIX G